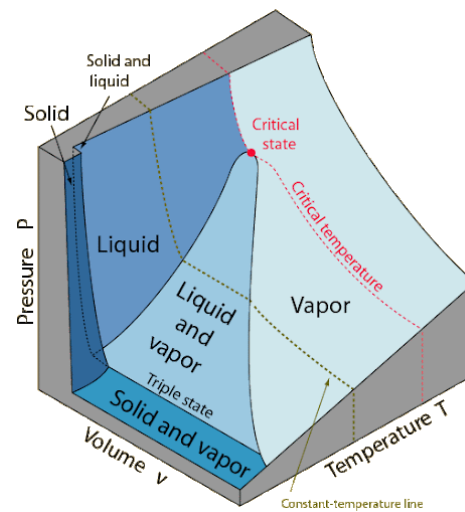
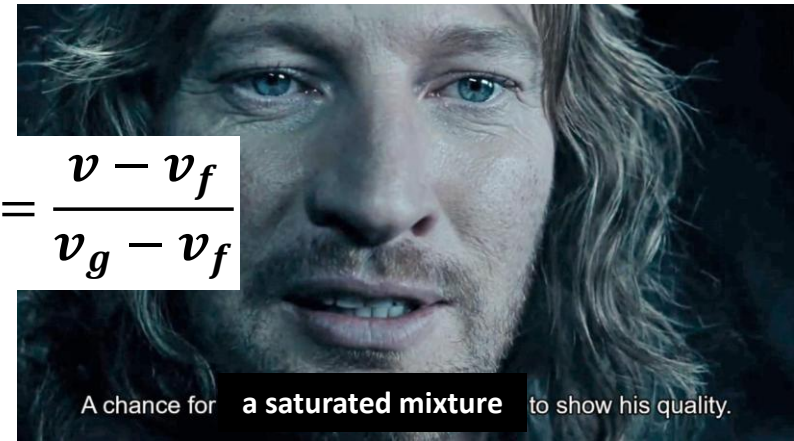


TEACHER: THERE ARE 3 STATES OF MATTER

THAT ONE KID:



$$x = \frac{v - v_f}{v_g - v_f}$$



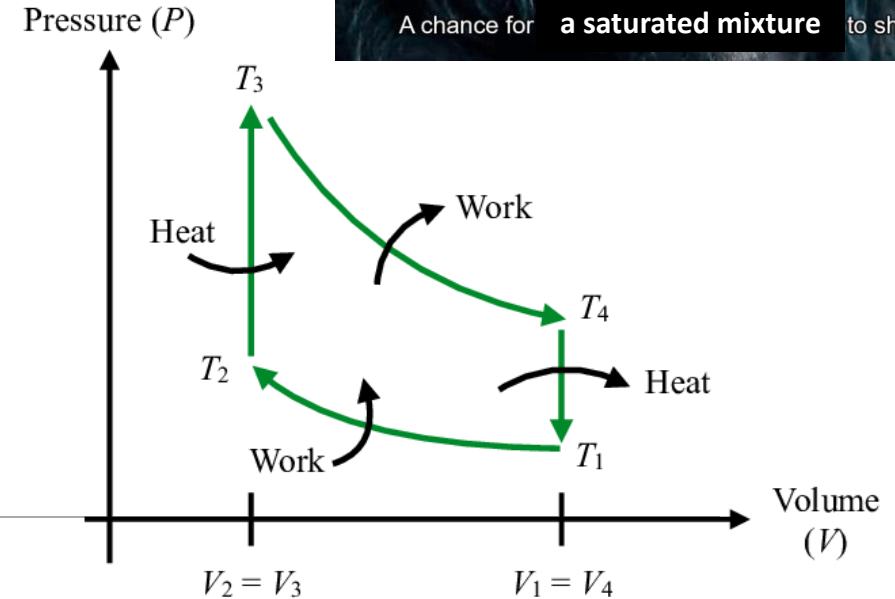
# ME 2134

## Thermodynamics

INSTRUCTOR: DR. JUSTIN DUBIK

SPRING 2026 – WEEK 2

CYCLES, EVALUATING PROPERTIES, P-V-T RELATIONS, PHASE CHANGES



# Why is thermodynamics useful?

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# Introduction to cycles

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# Change in properties

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# A reminder: states vs. processes

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# Cycle energy balance

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# Where is this work coming from?

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# Cycles example problem

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- A gas within a piston-cylinder assembly undergoes a cycle consisting of three processes
- Process 1-2: increase in pressure from  $P_1 = 1$  bar to  $P_2 = 3$  bar under constant volume  $V_1 = V_2 = 1 \text{ m}^3$ ,  $U_2 - U_1 = 400 \text{ kJ}$
- Process 2-3: constant pressure expansion to  $V_3 = 2 \text{ m}^3$
- Process 3-1: Adiabatic compression, with  $W_{31} = -1120 \text{ kJ}$
- What's the net work for the cycle?
- What are the heat transfers for each process?



# Cycles: how we use/generate work

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# Tracking change in energy (aka how to solve 1<sup>st</sup> law problems)

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# State principle

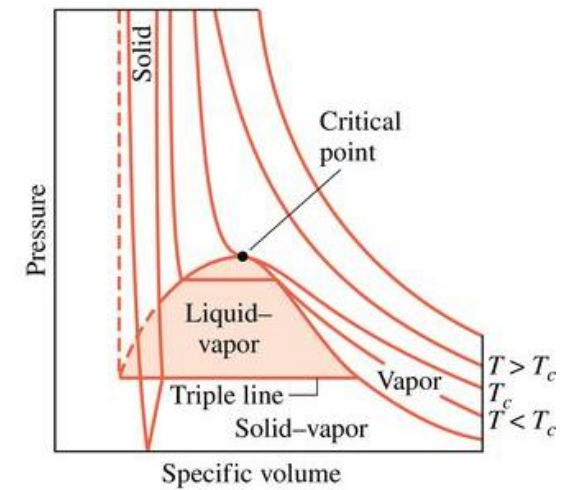
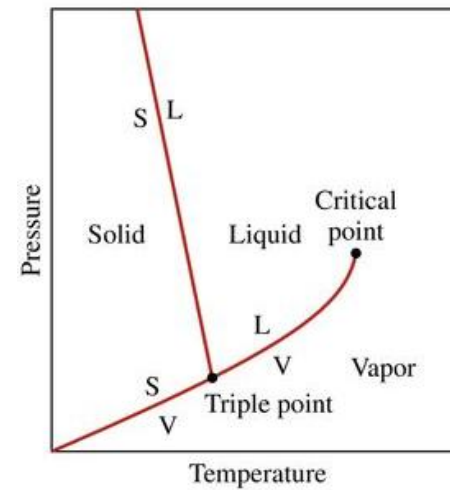
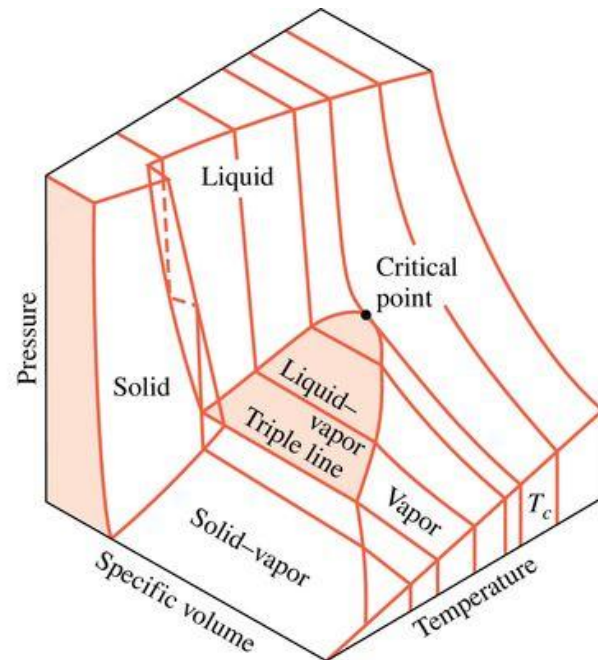
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# State principle (cont.)

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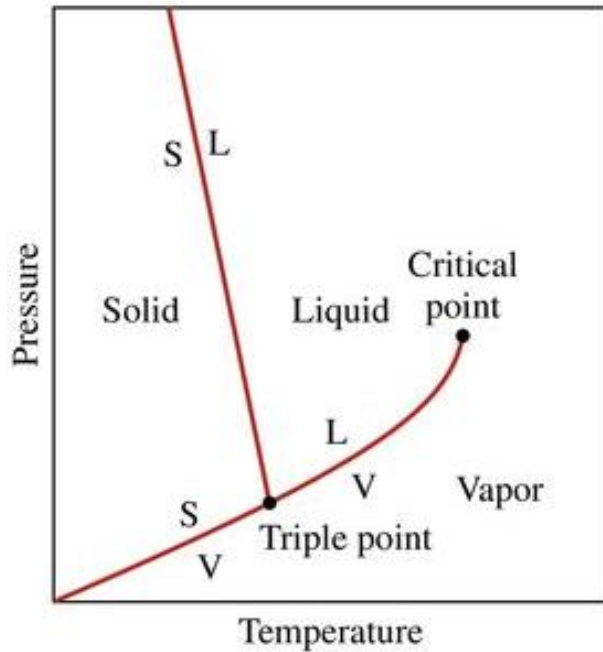
(Note: the state principle is a postulate)

# $P$ - $v$ - $T$ relationship

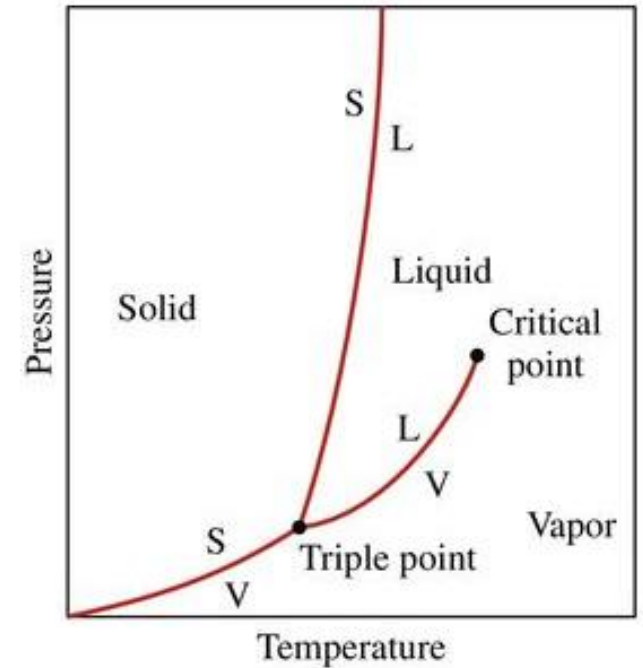


# $P$ - $T$ relationship

Expands on freezing:

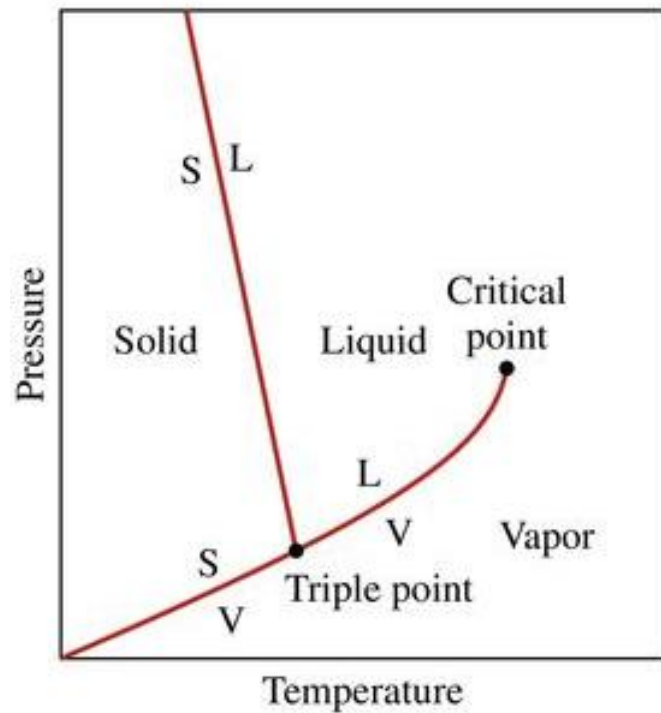


Contracts on freezing:



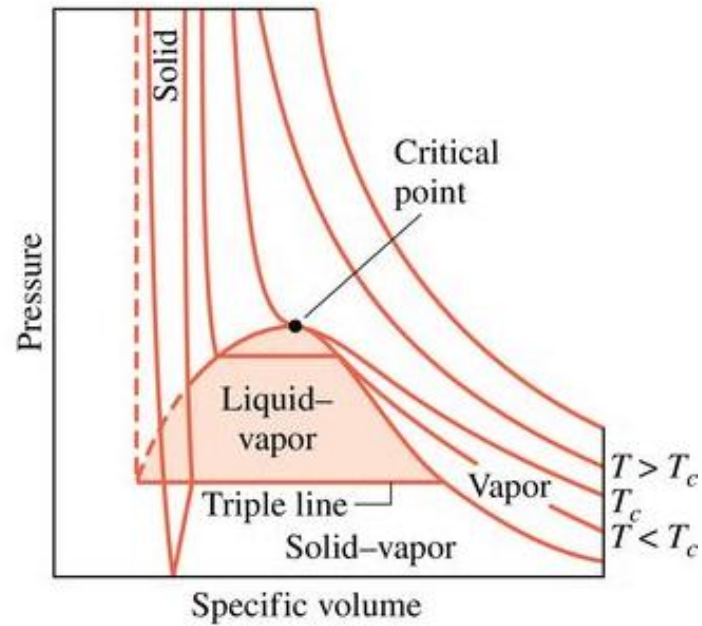
# $P$ - $T$ relationship (cont.)

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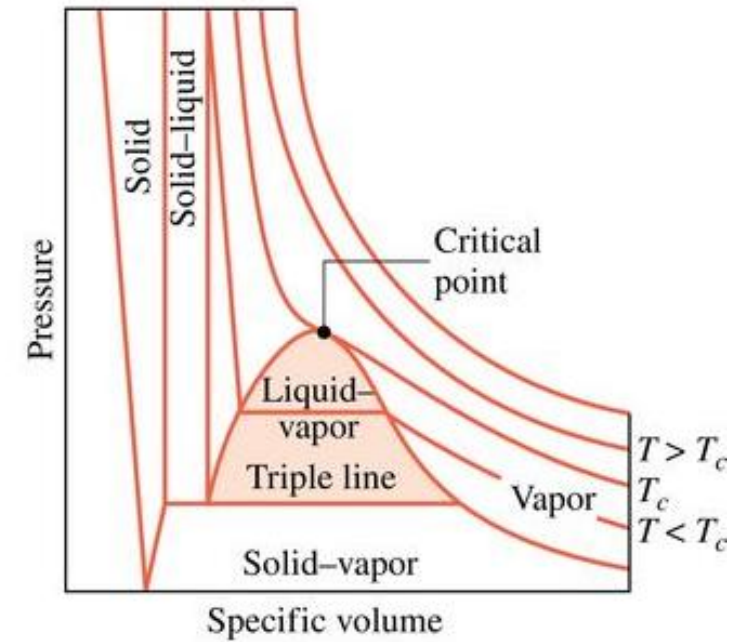


# $P$ - $v$ relationship

Expands on freezing:

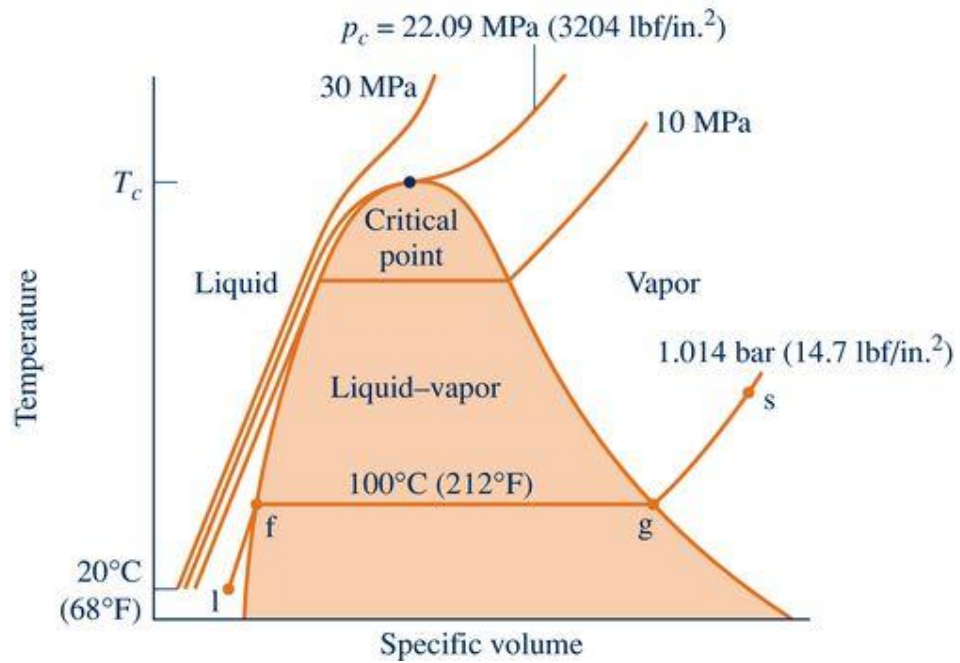


Contracts on freezing:



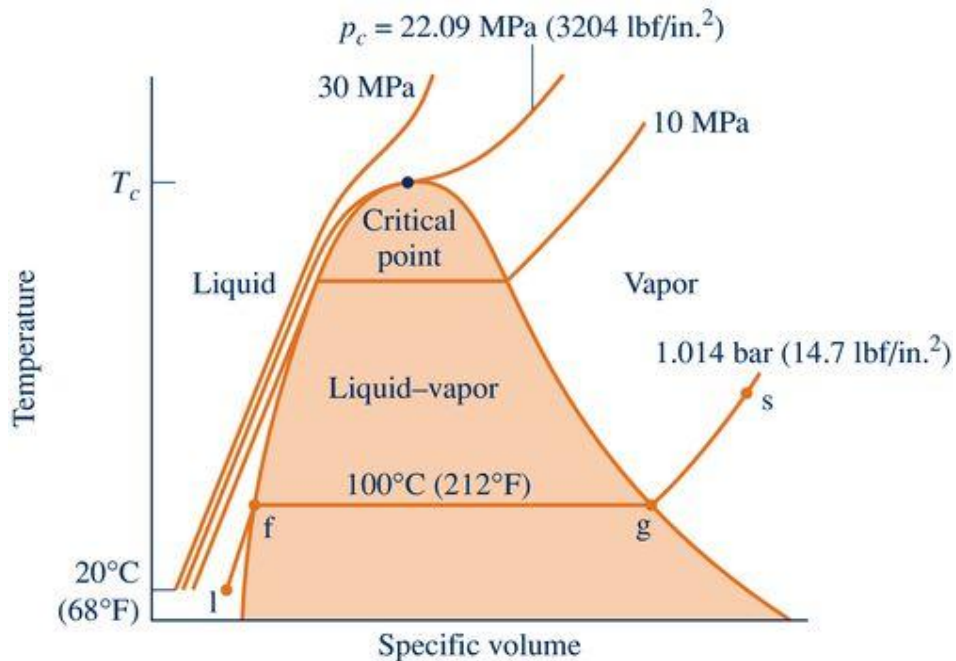


# $T$ - $v$ relationship



Using  $P, v, T$  relationships to fix state (find missing info, get  $u$ )

## How can we fill the following table?

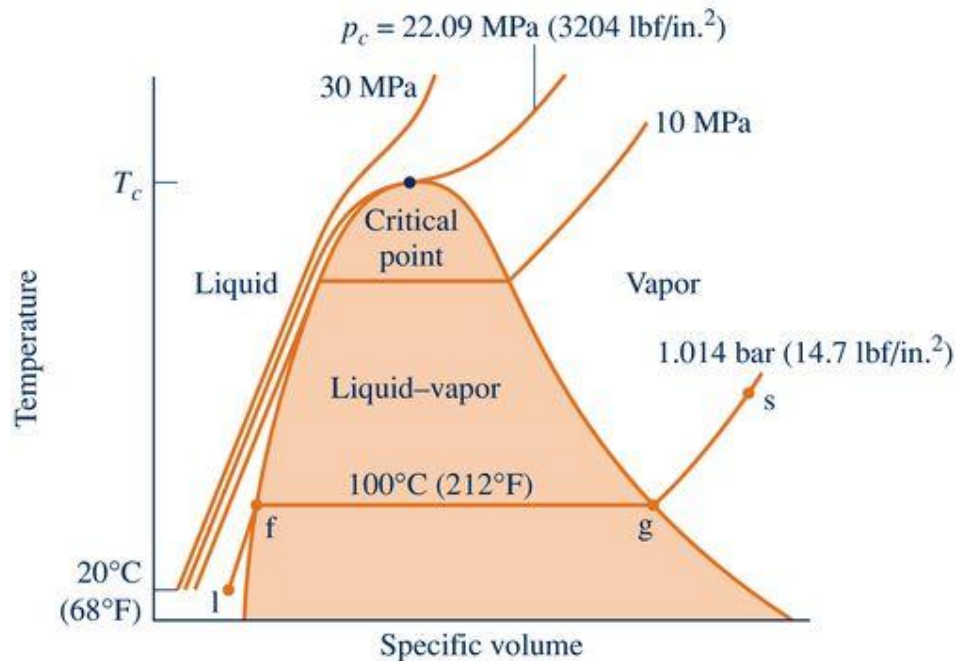


| <i>H<sub>2</sub>O</i> |                               |               |                  |                 |
|-----------------------|-------------------------------|---------------|------------------|-----------------|
| <i>P</i> (bar)        | <i>v</i> (m <sup>3</sup> /kg) | <i>T</i> (°C) | <i>u</i> (kJ/kg) | Phase           |
|                       |                               | 240           |                  | Saturated Vapor |
| 3                     | 0.907                         |               |                  |                 |
|                       | 0.1                           | 200           |                  |                 |
| 25                    |                               | 100           |                  |                 |

# Evaluating properties (cont.)

Using  $P$ ,  $v$ ,  $T$  relationships to fix state (find missing info, get  $u$ )

Tables A-2 to A-5 for water in liquid-gas states



## 15. Appendix to Tables in SI Units (Tables A-1 to A-27)

15.2 Properties of saturated water (liquid-vapor): temperature table (Table A-2)

15.3 Properties of saturated water (liquid-vapor): pressure table (Table A-3)

15.4 Properties of superheated water vapor (Table A-4)

15.5 Properties of compressed liquid water (Table A-5)

# Evaluating properties using tables (e.g. Table A-2)

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| Pressure<br>Conversions:<br>1 bar = 0.1 MPa<br>1 bar = 10 <sup>2</sup> kPa |               | Specific Volume<br>m <sup>3</sup> /kg |                        | Internal Energy<br>kJ/kg |                        | Enthalpy<br>kJ/kg       |                   |                        | Entropy<br>kJ/kg · K    |                        |             |
|----------------------------------------------------------------------------|---------------|---------------------------------------|------------------------|--------------------------|------------------------|-------------------------|-------------------|------------------------|-------------------------|------------------------|-------------|
| Temp.<br>°C                                                                | Press.<br>bar | Sat.<br>Liquid<br>$v_f \times 10^3$   | Sat.<br>Vapor<br>$v_g$ | Sat.<br>Liquid<br>$u_f$  | Sat.<br>Vapor<br>$u_g$ | Sat.<br>Liquid<br>$h_f$ | Evap.<br>$h_{fg}$ | Sat.<br>Vapor<br>$h_g$ | Sat.<br>Liquid<br>$s_f$ | Sat.<br>Vapor<br>$s_g$ | Temp.<br>°C |
| .01                                                                        | 0.00611       | 1.0002                                | 206.136                | 0.00                     | 2375.3                 | 0.01                    | 2501.3            | 2501.4                 | 0.0000                  | 9.1562                 | .01         |
| 4                                                                          | 0.00813       | 1.0001                                | 157.232                | 16.77                    | 2380.9                 | 16.78                   | 2491.9            | 2508.7                 | 0.0610                  | 9.0514                 | 4           |
| 5                                                                          | 0.00872       | 1.0001                                | 147.120                | 20.97                    | 2382.3                 | 20.98                   | 2489.6            | 2510.6                 | 0.0761                  | 9.0257                 | 5           |
| 6                                                                          | 0.00935       | 1.0001                                | 137.734                | 25.19                    | 2383.6                 | 25.20                   | 2487.2            | 2512.4                 | 0.0912                  | 9.0003                 | 6           |

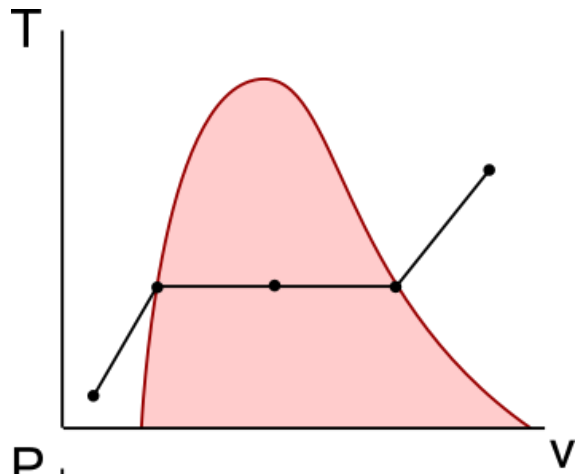
# More properties tables

| $T$<br>°C                                                                                | $v$<br>m³/kg | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m³/kg                                                                             | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K |
|------------------------------------------------------------------------------------------|--------------|--------------|--------------|----------------|------------------------------------------------------------------------------------------|--------------|--------------|----------------|
| $p = 0.06 \text{ bar} = 0.006 \text{ MPa}$<br>( $T_{\text{sat}} = 36.16^\circ\text{C}$ ) |              |              |              |                | $p = 0.35 \text{ bar} = 0.035 \text{ MPa}$<br>( $T_{\text{sat}} = 72.69^\circ\text{C}$ ) |              |              |                |
| Sat.                                                                                     | 23.739       | 2425.0       | 2567.4       | 8.3304         | 4.526                                                                                    | 2473.0       | 2631.4       | 7.7158         |
| 80                                                                                       | 27.132       | 2487.3       | 2650.1       | 8.5804         | 4.625                                                                                    | 2483.7       | 2645.6       | 7.7564         |
| 120                                                                                      | 30.219       | 2544.7       | 2726.0       | 8.7840         | 5.163                                                                                    | 2542.4       | 2723.1       | 7.9644         |
| 160                                                                                      | 33.302       | 2602.7       | 2802.5       | 8.9693         | 5.696                                                                                    | 2601.2       | 2800.6       | 8.1519         |
| 200                                                                                      | 36.383       | 2661.4       | 2879.7       | 9.1398         | 6.228                                                                                    | 2660.4       | 2878.4       | 8.3237         |
| 240                                                                                      | 39.462       | 2721.0       | 2957.8       | 9.2982         | 6.758                                                                                    | 2720.3       | 2956.8       | 8.4828         |
| 280                                                                                      | 42.540       | 2781.5       | 3036.8       | 9.4464         | 7.287                                                                                    | 2780.9       | 3036.0       | 8.6314         |
| 320                                                                                      | 45.618       | 2843.0       | 3116.7       | 9.5859         | 7.815                                                                                    | 2842.5       | 3116.1       | 8.7712         |
| 360                                                                                      | 48.696       | 2905.5       | 3197.7       | 9.7180         | 8.344                                                                                    | 2905.1       | 3197.1       | 8.9034         |
| 400                                                                                      | 51.774       | 2969.0       | 3279.6       | 9.8435         | 8.872                                                                                    | 2968.6       | 3279.2       | 9.0291         |
| 440                                                                                      | 54.851       | 3033.5       | 3362.6       | 9.9633         | 9.400                                                                                    | 3033.2       | 3362.2       | 9.1490         |
| 500                                                                                      | 59.467       | 3132.3       | 3489.1       | 10.1336        | 10.192                                                                                   | 3132.1       | 3488.8       | 9.3194         |

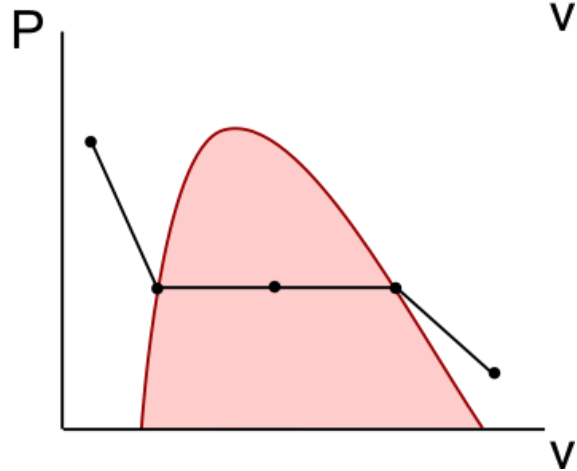
| $T$<br>°C                                                                               | $v \times 10^3$<br>m³/kg | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v \times 10^3$<br>m³/kg                                                                | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K |
|-----------------------------------------------------------------------------------------|--------------------------|--------------|--------------|----------------|-----------------------------------------------------------------------------------------|--------------|--------------|----------------|
| $p = 25.0 \text{ bar} = 2.5 \text{ MPa}$<br>( $T_{\text{sat}} = 223.99^\circ\text{C}$ ) |                          |              |              |                | $p = 50.0 \text{ bar} = 5.0 \text{ MPa}$<br>( $T_{\text{sat}} = 263.99^\circ\text{C}$ ) |              |              |                |
| 20                                                                                      | 1.0006                   | 83.80        | 86.30        | .2961          | .9995                                                                                   | 83.65        | 88.65        | .2956          |
| 40                                                                                      | 1.0067                   | 167.25       | 169.77       | .5715          | 1.0056                                                                                  | 166.95       | 171.97       | .5705          |
| 80                                                                                      | 1.0280                   | 334.29       | 336.86       | 1.0737         | 1.0268                                                                                  | 333.72       | 338.85       | 1.0720         |
| 100                                                                                     | 1.0423                   | 418.24       | 420.85       | 1.3050         | 1.0410                                                                                  | 417.52       | 422.72       | 1.3030         |
| 140                                                                                     | 1.0784                   | 587.82       | 590.52       | 1.7369         | 1.0768                                                                                  | 586.76       | 592.15       | 1.7343         |
| 180                                                                                     | 1.1261                   | 761.16       | 763.97       | 2.1375         | 1.1240                                                                                  | 759.63       | 765.25       | 2.1341         |
| 200                                                                                     | 1.1555                   | 849.9        | 852.8        | 2.3294         | 1.1530                                                                                  | 848.1        | 853.9        | 2.3255         |
| 220                                                                                     | 1.1898                   | 940.7        | 943.7        | 2.5174         | 1.1866                                                                                  | 938.4        | 944.4        | 2.5128         |
| Sat.                                                                                    | 1.1973                   | 959.1        | 962.1        | 2.5546         | 1.2859                                                                                  | 1147.8       | 1154.2       | 2.9202         |

# A summary of the various phases

---



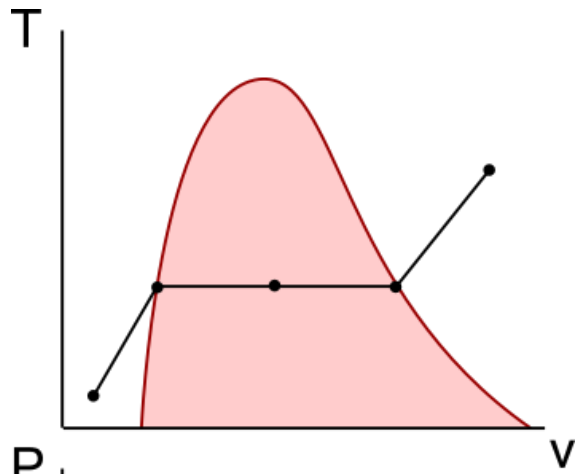
1.



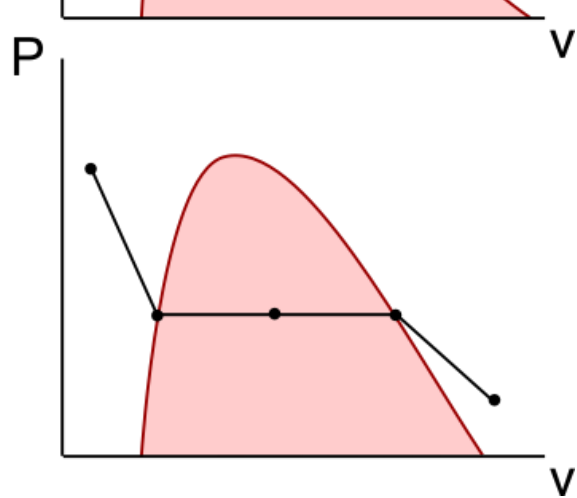
5.

# A summary of the various phases (cont.)

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2.

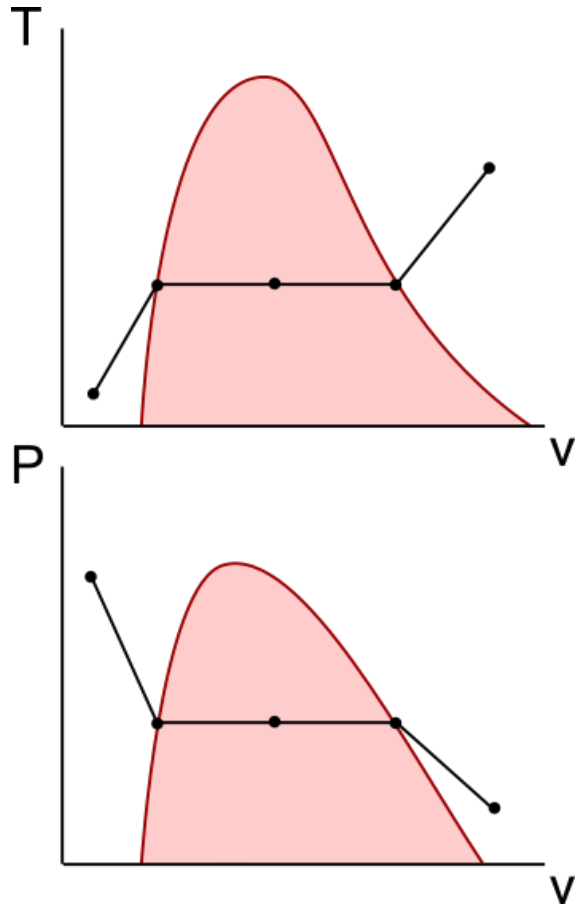


4.

# Determining properties - mixed phases

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3.



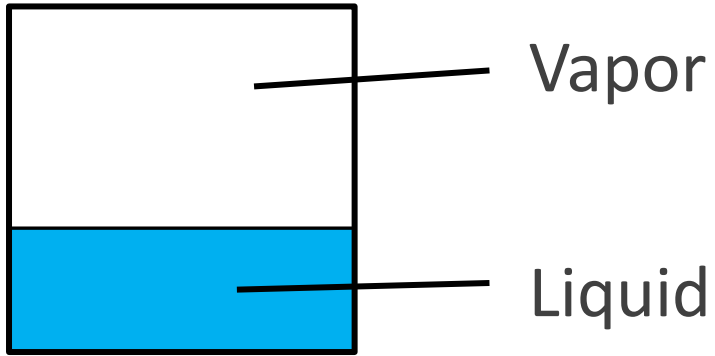


# Determining properties - quality

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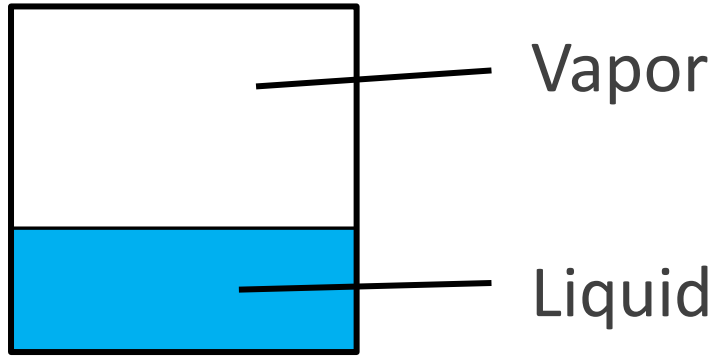
## Determining properties – quality (cont.)

---



# Determining properties – quality (cont.)

---



# Determining properties – the table example

---

| <i>H<sub>2</sub>O</i> |                          |          |             |                 |
|-----------------------|--------------------------|----------|-------------|-----------------|
| $P$ (bar)             | $v$ (m <sup>3</sup> /kg) | $T$ (°C) | $u$ (kJ/kg) | Phase           |
|                       |                          | 240      |             | Saturated Vapor |
| 3                     | 0.907                    |          |             |                 |
|                       | 0.1                      | 200      |             |                 |
| 25                    |                          | 100      |             |                 |

For the 3<sup>rd</sup> row (determining  $u$ )

---

# Determining properties – algorithm

---

If given T and P

- Go to T or P table (depending on which value is “cleaner”)
- In T table (Table A-2)
  - If  $P < P_{sat}$ : superheated vapor, use Table A-4
  - If  $P > P_{sat}$ : compressed liquid, use Table A-5
    - ~incompressible: often assume  $u \approx u_f@P_{sat}$ ,  $v \approx v_f@P_{sat}$
  - If  $P = P_{sat}$ : saturated mixture, stay in Table A-2
    - Will need  $u$ ,  $v$ ,  $x$ , etc. to determine if it's a vapor, liquid, or mixture

# Determining properties – algorithm (cont.)

---

If given T and P

- Go to T or P table (depending on which value is “cleaner”)
- In P table (Table A-3)
  - If  $T > T_{sat}$ : superheated vapor, use Table A-4
  - If  $T < T_{sat}$ : compressed liquid, use Table A-5
    - ~incompressible: often assume  $u \approx u_f@T_{sat}$ ,  $v \approx v_f@T_{sat}$
  - If  $T = T_{sat}$ : saturated, stay in Table A-3
    - Will need  $u$ ,  $v$ ,  $x$ , etc. to determine if it's a vapor, liquid, or mixture

# Determining properties – algorithm (cont.)

---

If given  $T$  or  $P$  (not both) and  $u$  or  $v$  (or  $h$  or  $s$ )

- Go to  $T$  or  $P$  table (depending on which is given)
- In either table:
  - If  $u > u_g$ : superheated vapor, use Table A-4
  - If  $u < u_f$ : compressed liquid, use Table A-5
  - If  $u_f < u < u_g$ : saturated mixture, stay in Table A-2/A-3
    - Can use  $u$  or  $v$  to determine  $x$
  - If  $u = u_f$  (or  $u = u_g$ ): saturated liquid (or saturated vapor)
    - Use corresponding values directly in Table A-2/A-3



# Value not in the table? Linear interpolation

---

Consider water at  $P = 0.35$  bar,  $T = 300$  °C: what is  $v$ ?